

- 1- Question below of the textbook (2-14). Hint: Write the chemical reaction, mole numbers of the products are given in the table, obtain mole numbers of reactants and use the table below. (Solution: $T_p=3500$ K)

Find the flame temperature of the products of combustion of hydrogen and oxygen whose composition and average specific heats are given in the table below. The reactants enter the adiabatic combustion chamber at 250 K. In this table, $\bar{C}_{pp}$ is the average molar specific heat of the component between 298 K and the flame temperature, $\bar{C}_{pr}$ is the average molar specific heat of the component between 250 K and 298 K, and $\bar{h}_f^0$ is the heat of formation at 298 K in kJ/kmol.

Component	Moles	$\bar{C}_{pp}$	$\bar{C}_{pr}$	Q_f
O ₂	0.1008	36.3	32.3	0
H ₂	0.3170	32.9	29.6	0
O	0.054	20.9		+245,143
H	0.109	20.9		+216,093
OH	0.233	33.5		+ 41,870
H ₂ O	1.512	48.27		-241,827

- 2- In the combustion chamber of a rocket with gaseous hydrogen as fuel and gaseous oxygen as oxidizers, the inlet temperature is 298 Kelvin, and the pressure is 210 atm. By changing the oxidizer-to-fuel mass ratio from 1 to 8, draw the curve of **a-** the temperature of the products with the oxidizer-to-fuel ratio and **b-** the mole of unburned hydrogen, OH, H₂O, H, O, O₂ with the oxidizer-to-fuel ratio. Use CEA software.